Oliver Li

Skills: Go, Scala, TypeScript, Python, Vue.js, Java, C, Computer Networks, Git, Linux, LaTeX
Interests: Computing History, Board Games, Personal Finance, General Aviation, Operating Systems
Languages: English (native), Chinese (near-native), German (entry level, A2)
Location: New York City

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Experience

Datadog - Software Engineer II, New York, NY — Jun 2024 - Present

- Designed and implemented the public App Builder API, with $O(n)$ JSON Schema validation of app definitions via a cgo compatibility layer calling a Rust validation library from Go; the API release enabled the official Terraform provider for App Builder
- Designed App Builder's contributor permission model and an optional editor review flow (analogous to GitHub pull requests) requiring a second editor to approve changes before publication, giving teams native SDLC controls over app changes without requiring external version control
- Refactored error handling across App Builder, Datastores, and Forms around a typed error abstraction that enforced actionable end-user messages (e.g. field-level validation failures rather than opaque 5xx responses) and prevented internal implementation details from surfacing in API responses

Nuvo - Software Engineer, New York, NY — Jan 2023 - May 2024

- Designed and implemented Nuvo's integration with NetSuite, with bidirectional syncing of customer and accounts receivable information; built an accounts receivable dashboard for credit managers to monitor trade credit exposure
- Shipped a task queue service using the ARQ library with Redis; implemented a dead letter queue, static type checking support, and a full-featured dashboard, enabling faster development and less latency in the core application
- Built an idempotent batch backfill pipeline for importing pre-existing approved credit applications on customer onboarding; enforced data integrity via EIN-to-customer matching and a unique index on customer-supplier pairs, and deliberately rate-limited imports to throttle downstream email notifications and catch errors before propagation

Stripe - Software Engineer, Seattle, WA — Jun 2022 - Nov 2022

- Implemented transaction holds and releases in Stripe's next-generation Java stack as part of the credit card clearing team, ensuring funds settled on time and in full
- Investigated and resolved merchant-facing race conditions in the clearing service; reduced the p99 latency of a major Kafka consumer by 85%
- Automated repetitive oncall toil, including a batch script surfacing transactions requiring manual partner intervention to Mastercard and other network partners

Education

B.S., Electrical & Computer Engineering - Carnegie Mellon University — Aug 2019 – May 2022

- Teaching assistant for 15-441 (Computer Networks); led the TCP project, covering 3-way handshake, reliable transmission, and flow control
- Other courses completed include 18-349 (Embedded Systems), 18-344 (The Hardware/Software Interface), 18-330 (Computer Security), and their prerequisites. Achieved an overall GPA of 3.85
- Graduated 1 year early